

Shawn Yong

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EDUCATION

University of Delaware

Bachelor of Mechanical Engineering (BME); Major: Mechanical Engineering, Minor: Math; GPA: 3.5 **May 2025**
Newark, DE

EXPERIENCE

Leidos

Distribution Engineer

May 2026 – Present
Wilmington, North Carolina

- Designed AutoCad distribution maps & BOMs with 85% accuracy to execute upgrades on power distribution circuits
- Applied distribution standards to outdated utility poles and distribution equipment (transformers, fuses, down guys)
- Conducted field surveys, measuring pole dimensions, photographing distribution equipment with clear lines of sights, and documenting unrecorded conductor circuitry to ensure 100% GIS map accuracy
- Onboarded new team members with important design skills and techniques, helping them reach 60% accuracy on distribution maps and BOMs while meeting metrics within 2 months
- Coordinated with power utility seniors and permitting departments to calculate optimal fusing sizes, sufficient power delivery, and alignment on permissible power distribution construction sites

VisionNav Robotics

Project Engineer

February 2026 - May 2026
Atlanta, Georgia

- Analyzed operational workflows and warehouse layouts to optimize autonomous forklift routes, reducing cycle times and congestion in high-traffic areas
- Trained clients on system operation, safety protocols, troubleshooting procedures, and error-preventative measures
- Retrofitted robotic forklifts with sensors to improve spatial sensing inside irregular pallet holes
- Collaborated with IT and the customers to identify technical issues, and implement process improvements to enhance service standards and project execution efficiency
- Developed and documented (SOPs) for installation, operation, and maintenance of VisionNav systems to improve consistency across project deployments

Center for Composite Materials (CCM)

Research Assistant

June 2024 - May 2025
Newark, Delaware

- Investigated the deconsolidation of thermoplastic composites, analyzing the effect of varied fiber materials within a consistent plastic matrix
- Performed tensile and compression tests while strictly adhering to ASTM testing standards for each unique specimen
- Characterized composite properties using DSC, confocal microscopy, TGA, microCT scans, and deconsolidation tests
- Demonstrated visualized data by generating standardized, clear images and graphs for weekly presentations to peers and senior professionals, facilitating progress updates and technical feedback.
- Presented thermoplastic composite deconsolidation research to academic and industry experts at multiple symposiums
- Awarded 1st place at the CCM AIM Undergraduate Poster Competition for "outstanding research and exemplary presentation skills"

PROJECTS

Pill Filling Machine

3D Printing, Laser Cutting, Motorizing

- Developed and wired sensor-driven control systems using Arduino micro controllers to manage stepper motors for automated pill dispensing and cap-screwing machinery.
- Generated curved cap chute laser cuts by illustrating a live kerf pattern and tolerancing joints for a precise assembly
- Optimized project parts by strategically orienting 3D-printed grain lines during 3D modeling and printing to maximize structural strength specific to the intended loading
- Optimized motor and sensor control algorithms within Arduino IDE to enhance prototype performance while minimizing heat, noise, and vibration
- Written user manuals for final prototypes, including detailed component labeling, circuit diagrams, clear step-by step assemblies and usage instructions

Under Armour

3D Printing, CNC Machining, Motorizing

- Worked with a team of 3 to design a shoe squeak tester with adjustable surfaces and loading to analyze different sole patterns and compounds
- Programmed CNC machine with ProtoTrak SMX to fabricate aluminum frame parts that withstand loads up to 200lbs
- Wired up a sturdy motorized rotary table controlled by an easy to use motor controller with programmable test algorithms
- Conducted multiple tests with different sole samples & surfaces and presented findings to the UA materials engineering department

TECHNICAL SKILLS

Software: SolidWorks, Fusion 360, OnShape, ProtoTrak SMX, Python, MATLAB, C++, Arduino IDE, TinkerCAD, Microsoft Office Suite, Adobe Illustration, Finite Element Analysis (FEA)

Machines: 3D printers, Laser cutters, Arduino Boards, Hand Tools, Milling Machines, CNC, Computers, Sewing Machine, DSC, Confocal Microscope,